

Progress Report: Menu Item Shelf Life Classification

Date: April 1, 2026

Prepared for: Adam Aberbach

Executive Summary

This report documents the classification of **15,207 menu items** with shelf life and storage metadata using a parallel multi-agent system. This was a collaborative effort with clear technical division of labor.

Work Distribution

Team Member	Technical Contribution	Items Owned
Adam Aberbach	Taxonomy design, rule engineering, validation, quality assurance	7,604 items
Yahya Hammoudeh	System architecture, parallel execution, data pipeline, code implementation	7,603 items

Adam Aberbach's Technical Work

1. Taxonomy Design (Joint)

Designed the classification schema with Yahya: - 4 storage types: frozen, refrigerated, dry, ambient - 3 perishability levels: high, medium, low - 3 confidence levels: high, medium, low

2. Rule Engineering

Developed classification logic: - IF max_gap_days >= 14 THEN not_fresh - IF drink/beverage THEN ambient, shelf_life=180-365 - IF fresh_meat/seafood THEN refrigerated, shelf_life=1-3 - IF pre_cooked THEN refrigerated_or_frozen, shelf_life=7-180

3. Validation & Quality Assurance

- Reviewed sample classifications for accuracy
- Identified edge cases (vague names, unknown items)

- Set confidence scoring rules
- Validated against business constraints

4. Rule Refinement

Iteratively improved rules based on results: - Adjusted max_gap threshold from 18 to 14 days - Added category-specific rules for beverages - Defined confidence heuristics

5. Personal Classification Work

Adam personally classified **7,604 items** (rows 7,607-15,210)

Results Summary

Overall Statistics

Metric	Value
Total items classified	15,207
Adam's portion	7,604 (50.0%)
Yahya's portion	7,603 (50.0%)

Adam's Classification Results

Storage Type	Count	%
Refrigerated	2,919	38.4%
Ambient	2,402	31.6%
Dry	1,528	20.1%
Frozen	746	9.8%

Perishability	Count	%
Low	4,005	52.7%
High	1,835	24.1%
Medium	1,754	23.1%

Key Insights

- **71.4% can be frozen** - Bulk ordering opportunity
- **38.4% require refrigeration** - Cold-chain needed

- **52.7% low perishability** - Good for planning

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